A Novel Cleaning Approach of Environmental Sensing Data Streams

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***Abstract*—With recent widespread usage of state-of-the-art technology (e.g., various mobile devices), environmental sensing is getting popular. The sensors used for sensing are small and due to the mobility they become more error-prone, which results in data corruption or loss from sensor. Therefore, cleaning of the sensed data is of high importance to recover the lost or corrupted data. In this paper, we propose a novel data cleaning mechanism to ensure better accuracy in environmental sensing applications. Based on the sensed data and the context relationship of each sensor, we update the credibility (or alternatively reliability) of the sensed data. We consider mobility pattern of the mobile sensor nodes while selecting the candidate sensor nodes for data stream cleaning. Through simulations, we evaluate the performance of our proposed approach. We compare our proposed sensor data stream cleaning approach with Influence Mean Cleaning (IMC) (a recent algorithm in data stream cleaning) and Mean-based cleaning. Simulation results show up to 24% reduction in root mean square error (RMSE) over IMC and up to 30% over Mean-based cleaning.**

I. Introduction

In mobile Wireless Sensor Networks (mWSN), uncertainty is a common phenomenon, where nodes change their positions rapidly and unpredictably. Although, reliability and accuracy is of utter importance in many sensor applications, it is often difficult to ensure these properties. For example: energy scarcity and frequent movement of the sensor nodes often contributes to imprecise or dirty data [1]. To recover the lost data, it is needed to find out the correlated sensor data that can be used in the prediction method. Data cleaning deals with missing values, noisy data, inconsistent data etc. [2]. Many recent research works have focused on using post-processing data cleaning at the server end. There also exist some efforts on using online data stream cleaning method in mWSN [3]. However, not much work has focused on how sensor context can be used in sensor selection for data cleaning.

In environmental sensing, different sensors are deployed to sense various environmental properties (e.g., humidity, temperature, ozone) [4]. The sensors are mounted on top of different vehicles (e.g., bicycle, tram) or carried by a human being; which change their positions very frequently and unpredictibly. Therefore, the network structure for these sensor nodes changes dynamically, making it imperative to consider the mobility pattern for sensor data.

In [5], the authors have proposed IMC method for environmental sensing based on incrementally adjusted reliability of individual sensor. The reliability measurement does not require prior hardware knowledge. With the advance of time, they incrementally adjusted the reliability of each sensor based on their sensing data accuracy. In our paper, we considered not only the sensed data value, but also the context of the sensor along with mobility pattern of the mobile sensors. With the combination of both of these comparisons, our method performs better cleaning and helps in ensuring higher accuracy in environmental sensing.

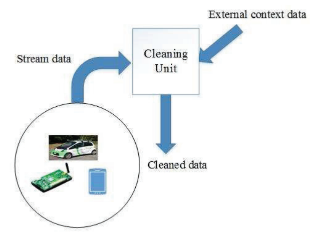

Fig. 1. Cleaning Unit

II. Architecture

In our system, we calculate sensor credibility according to the historical sensing performance as well as considering the context of the sensor. The word credibility refers to the quality of being believed or accepted as true. Sensor credibility is the ratio of the number of times it senses data correctly and the total number of sensing performed by that sensor during the desired time period [5]. For all the data samples that a sensor senses during a particular time window, the difference is compared with the predicted value. The mean value of all data samples from spatially correlated sensors is calculated and used as predicted value. If the difference is in a tolerable range, the sensed data is considered correct. The sensor with higher credibility has higher impact during correlation.

We calculate context credibility by comparing the context value of the sensor with the neighboring sensors co-located during a similar time window. If the difference is within a tolerable limit, then higher weight is being assigned (w_2 in Algorithm 1). By incorporating context, we bring dynamism in area classification. Two nodes located nearby in the same square region get different levels of importance due to their context value (e.g., change in terrain elevation, height above

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Algorithm 1 Cleaning Algorithm

Input : sensor_id

block = region[sensor_id]
context_val $\leftarrow$ context[$sensor\_id$]
$S \leftarrow$ for each $s_i \in$ block during t_w
for all $S_i \in S$ **do**
  for all $r_i \in S_i$ **do**
    $w_1 \leftarrow$ Equation (1)
    $w_2 \leftarrow$ CalcContextwt(S_i,context_val)
    $w_3 \leftarrow$ MobilityAffinity(S_i,sensor_id)
  end for
  sum $\leftarrow r_i * w_1 * w_2 * w_3$
  weight $\leftarrow w_1 * w_2 * w_3$
  totalSum $\leftarrow$ totalSum + sum
  totalWeight $\leftarrow$ totalWeight + weight
end for
cleanedData $\leftarrow \dfrac{\text{totalSum}}{\text{totalWeight}}$

sea level). We have also analyzed the movement patterns of the sensor nodes. Higher weight is being assigned to the sensor nodes that tend to move in close correspondence with the sensor that needs data cleaning during the desired time window. Sensors that move in a group are given more importance.

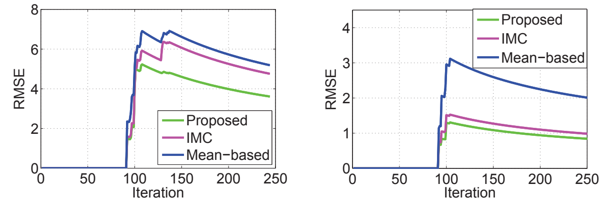
We have designed a centralized online data cleaning mechanism. We assume that there is a pre-processing mechanism that identifies the values that need to be cleaned. Our novel algorithm predicts the replacement value.

$$cred(data) = \begin{cases} 1, & |data - predicted| \leq threshold_d, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For all data of a sensor, the three weight values (w_1, w_2, w_3) are calculated by calling eqn. 1, `CalcContextwt` and `MobilityAffinity` respectively. We calculate the product of these three weight values and assign it to $weight$. These three weight values control the significance of each sensed data value in the prediction of the clean data. The product of the weights and sensed data (r_i) is stored in sum. The variables $totalSum$ and $totalWeight$ are used for the weighted moving average based $cleanedData$ calculation.

III. Performance Evaluation

We simulated a scenario of 10 mobile sensors moving in an area of $200m * 200m$ using ns-2 and BonnMotion. The simulation duration was $1000s$. Some of the mobile nodes moved in nomadic mobility pattern and others followed random waypoint mobility. Our proposed algorithm (along with IMC and Mean-based cleaning) was tested on a dataset of Smart City project in Melbourne [6]. Terrain elevation was added as context information using API [7]. During each iteration, sensor nodes sensed the humidity of the region it is located in. The whole region is divided into 100 equally sized blocks each having dimension of $20m * 20m$. The readings having spatial position within the same block were considered as a singular group. We considered fixed and variable speed for the mobile nodes.



(a) (b)

Fig. 2. RMSE comparison with IMC and Mean-based for (a) Continuous walking of the nodes, (b) Variable movement speed.

We calculated RMSE and used it as a performance measurement criteria for our algorithm. RMSE is a standard metric to evaluate the accuracy of the prediction model. Fig. 2(a) depicts the case of all the mobile nodes moving continuously in walking speed (e.g., 2mph) and sensing the humidity values. High error values from all participating sensors in that region caused a sudden rise at the initial stage. In Fig. 2(b) we show RMSE comparison for nodes moving at variable speed. Even though the node speed varies, our algorithm can predict the missing value with lower error than the compared algorithms.

IV. Conclusions and Future Work

In this paper, we proposed a novel mechanism for cleaning environmental sensing data streams that considers not only the sensed value, but also the sensor context and movement affinity for data cleaning. Because of the low quality and mobility around various environments, the data received from the tiny sensors are error-prone. Our method is a weighted moving average based mechanism that can predict the missing data value more accurately even when there is higher fluctuation in the data streams. In the future, we would like to explore heterogeneous sensor context and the effect of varied semantics in mobile sensing.

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